

Day 70 – VLANs & Network Segmentation

Overview

Day 70 introduced **Virtual Local Area Networks (VLANs)** and demonstrated how a single physical switch can be divided into multiple logical networks.

In previous labs:

- Day 65: RIP Dynamic Routing
- Day 66: DHCP
- Day 67: DNS Concepts
- Day 68: NAT
- Day 69: ACL Security

Day 70 focused on **network segmentation**, a foundational enterprise networking concept used to separate users, departments, and resources.

The goal of this lab was to create multiple VLANs on a switch, assign switch ports to specific VLANs, and verify that devices in different VLANs could not communicate without routing.

Objectives

By the end of this lab, I was able to:

- Understand the purpose of VLANs.
 - Create VLANs on a Cisco switch.
 - Assign switch ports to VLANs.
 - Verify VLAN membership.
 - Understand broadcast domains.
 - Demonstrate VLAN isolation.
 - Understand the relationship between VLANs and enterprise networking.
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VLAN Fundamentals

What is a VLAN?

VLAN stands for:

Virtual Local Area Network

A VLAN allows multiple logical networks to exist on a single physical switch.

Instead of:

One Switch

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One Network

we can create:

One Switch

=

Multiple Networks

Why VLANs Exist

Without VLANs:

PC0

PC1

PC2

PC3

Same Switch

Same Network

All devices share the same broadcast domain.

With VLANs:

VLAN 10

PC0

VLAN 20

PC1

The switch separates traffic between groups.

This improves:

- Security
 - Performance
 - Organization
 - Scalability
-

Enterprise Example

A company may use:

VLAN 10 = HR

VLAN 20 = Sales

VLAN 30 = IT

Each department remains separated even while using the same switch infrastructure.

Lab Topology

A simple VLAN demonstration topology was created.

PC0

|

Fa0/1

|

Switch0

|

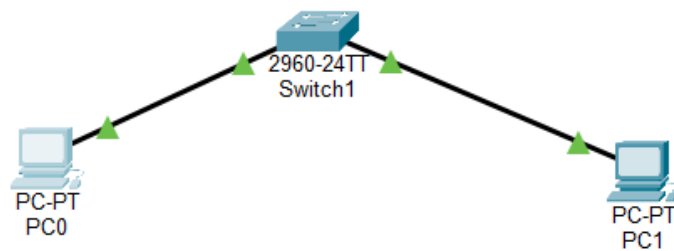
Fa0/2

|

PC1

Even though both PCs connect to the same switch, VLAN configuration separates them.

Screenshot #1 – Initial VLAN Lab Topology



Highlight

- PC0 connected to Fa0/1
 - PC1 connected to Fa0/2
 - Single switch
 - Physical connectivity established
-

VLAN Creation

Two VLANs were created.

enable

configure terminal

vlan 10

name HR

vlan 20

name SALES

Explanation

Created:

VLAN 10 = HR

and

VLAN 20 = SALES

These VLANs became separate logical networks inside the switch.

Screenshot #2 – VLAN Creation Commands

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name SALES
Switch(config-vlan)#
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#interface fa0/1
^
```

Highlight

vlan 10

name HR

vlan 20

name SALES

Port Assignment

PC0 was connected to:

Fa0/1

and assigned to:

VLAN 10

Configuration:

interface fa0/1

switchport mode access

switchport access vlan 10

PC1 was connected to:

Fa0/2

and assigned to:

VLAN 20

Configuration:

interface fa0/2

switchport mode access

switchport access vlan 20

Screenshot #3 – Port Assignment Commands

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#switchport mode access
      ^
```

Highlight

Fa0/1 → VLAN 10

Fa0/2 → VLAN 20

```
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#end
Switch#show vlan brief
%SYS-5-CONFIG_I: Configured from console by console
```

VLAN Verification

Verification command:

show vlan brief

Output:

10 HR active Fa0/1

20 SALES active Fa0/2

This verified that:

Fa0/1 belongs to VLAN 10

Fa0/2 belongs to VLAN 20

Screenshot #4 – VLAN Verification

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	HR	active	Fa0/1
20	SALES	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#interface fa0/1
^

Highlight

10 HR active Fa0/1

20 SALES active Fa0/2

PC Configuration

Static IP addresses were assigned.

PC0

IP Address:

192.168.10.10

Subnet Mask:

255.255.255.0

PC1

IP Address:

192.168.20.10

Subnet Mask:

255.255.255.0

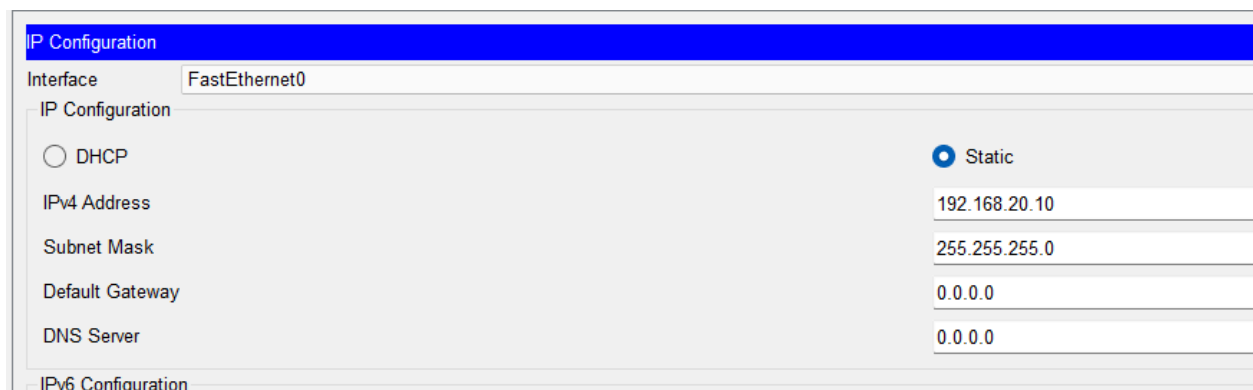
Screenshot #5 – PC0 IP Configuration



The screenshot shows the 'IP Configuration' window for the 'FastEthernet0' interface. The 'Static' radio button is selected. The configuration fields are as follows:

Field	Value
IPv4 Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

Screenshot #6 – PC1 IP Configuration



The screenshot shows the 'IP Configuration' window for the 'FastEthernet0' interface. The 'Static' radio button is selected. The configuration fields are as follows:

Field	Value
IPv4 Address	192.168.20.10
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

Below the IPv4 configuration, there is an 'IPv6 Configuration' section which is currently collapsed.

Connectivity Testing

From PC0:

ping 192.168.20.10

Result:

Failed

Why The Ping Failed

This failure actually proved the VLAN configuration was successful.

Although:

PC0

and

PC1

were connected to the same physical switch,

they were placed in:

Different VLANs

which means:

Different Broadcast Domains

Without a router performing Inter-VLAN Routing, communication is blocked.

Screenshot #7 – Failed Ping Test

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Highlight

Request Timed Out

or

Destination Host Unreachable

Key Lesson Learned

This lab demonstrated:

Same Switch

≠

Same Network

When VLANs are used.

The switch now behaves as though multiple separate switches exist.

Before VLANs

PC0

|

Switch

|

PC1

Communication Allowed

After VLANs

PC0

|

VLAN 10

VLAN 20

|

PC1

Communication Blocked

Cloud Engineering Connection

While AWS does not directly use Cisco VLANs, the concept appears through:

- VPC Segmentation
- Public Subnets
- Private Subnets
- Network Isolation
- Security Boundaries

Understanding VLANs helps explain why organizations separate workloads and departments into isolated network segments.

What I Learned

- VLANs create logical networks.
 - VLANs separate broadcast domains.
 - Multiple VLANs can exist on a single switch.
 - Ports can be assigned to specific VLANs.
 - Devices in different VLANs cannot communicate directly.
 - Inter-VLAN communication requires routing.
 - VLANs improve security and organization.
 - VLANs are foundational to enterprise networking.
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Reflection

Day 70 introduced VLANs and demonstrated how network segmentation works within a switch. I created VLAN 10 (HR) and VLAN 20 (SALES), assigned switch ports to each VLAN, and verified the configuration using `show vlan brief`. After assigning IP addresses to the PCs, cross-VLAN communication failed as expected, proving that the traffic was properly segmented. This lab helped reinforce the concept that one physical switch can support multiple logical networks and prepared me for future networking topics involving routing between VLANs and enterprise network design.
